

View Author Feedback

Paper ID

2635

Paper Title

ClearEval: A Data-driven Benchmark for Evaluating Large Language Models in Tissue Optical Clearing

AUTHOR FEEDBACK QUESTIONS

1. The space below is for both Rebuttal papers and early accepted papers. Instructions for Rebuttal papers: Your rebuttal is addressed to the Reviewers, Area Chairs, and Program Chairs. Reviewers will be able to see your rebuttals and have the option to change their ratings based on the rebuttal. If your paper is accepted, the rebuttal will be made public alongside the reviewer and meta-reviewer comments. The goal of the rebuttal is to inform the Reviewers and Area Chairs of major misunderstandings, in your opinion, in the reviewers' assessment, or of incorrect statements in the reviews. An effective rebuttal focuses only on major critiques. It is not helpful to try to address every minor point in the reviews. By prioritizing and focusing on the major concerns, and by grouping multiple reviewer comments that generally pertain to the same issue into a few major categories, you are demonstrating to the Reviewers and Area Chairs that you understand the high level messages that were provided in the reviews. Please summarize or rephrase the criticism before you address it, and clarify to which comment(s) you are responding. While the room for rebuttal is limited, if properly utilized by condensing the response down to the essentials, this is an effective way to let the Reviewers and Area Chairs know that you understood the reviewer's concerns and have valid answers to the questions raised in the reviews, or to establish that certain reviewer comments were false or unsubstantiated. An effective rebuttal addresses reviewers' criticisms by explaining where in the paper you had provided the requisite information, perhaps further clarifying it. Do not provide or promise new / additional experimental results, as the final decision is to be made based on the submitted manuscript. Note that the fundamental content of the paper, including experiments, data, analysis, etc. should not be changed. Do not promise to expand your paper to address all the questions raised by the reviewers, as you will not be able to change your article substantially, and in all likelihood you don't have sufficient room to add to the paper. These promises are likely not to be taken seriously. A good rebuttal is polite; being confrontational does not bring any added value to the paper. However, if you feel you have received a review that was not courteous, or made false or unsubstantiated arguments that you can succinctly refute, you should point this out. Remember that your rebuttal will be made public, so do not write anything you would not want to appear in print. Please note that including an external URL/website that contains additional information/explanation of your paper is prohibited. That is,

your rebuttal should be self-sufficient and within the limit of 4000 characters. External links will be removed. (See "Rebuttal Guide"

<https://conferences.miccai.org/2026/en/REBUTTAL-GUIDELINES.html>) Please enter your rebuttal below (4000 chars max) ----- For Provisional/Early Accepted paper: Please write your response to reviewer and meta-reviewer comments here. This is optional. You can use this to correct any misinterpretations or inaccuracies in the reviewer's findings. Your response will be made public alongside the reviewer comments. (4000 chars max)

We thank all reviewers for their constructive comments and the chairs for their efforts. To address specific concerns, we categorized our responses into: scope & motivation (R1, R2), benchmark construction (R2, R3), metric validity (R2, R3), human evaluation (R1, R2, R3), and baseline ablation & reproducibility (R2). Below, we respond point by point.

Scope & motivation (R1, R2). ClearEval is a pre-experimental checker for Tissue Optical Clearing (TOC). Rather than another LLM leaderboard, it evaluates whether a proposed protocol is complete, correct, and feasible before wet-lab execution. Specifically, although SOTA LLMs perform well on knowledge tasks, we aim to identify where they fail when turning knowledge into executable lab protocols. Moreover, establishing this text-reference-free benchmark serves as a necessary prerequisite for the future development of scientific agents dedicated to automated protocol design — which is why our evaluation focuses on foundation models.

Benchmark construction (R2, R3). ClearEval uses a two-stage design to test LLMs. Stage 1 (Knowledge) checks static TOC knowledge via multiple-choice questions, while Stage 2 (Application) checks practical protocol-design ability via open-ended questions. To ensure tasks are grounded in domain expertise, we built a shared database. Specifically, we collected 437 text excerpts from over 30 TOC articles and built a knowledge base covering tissue types, labeling sites, method properties, and compatibility. For Stage 1, we used a QA-generation workflow with the database to expand 70 template questions into 2589 MCQs. For Stage 2, combining 60 scientific tasks with our database, we generated 240 open-ended questions and added 20 real wet-lab challenges from experienced experts. After quality filtering, this resulted in 253 OEQs.

Metric validity and LLM-as-Judge calibration (R2, R3). ClearEval evaluates whether a protocol can actually work in the lab through our CCE (Completeness, Correctness, Effectiveness) framework, using LLM-as-a-Judge only where appropriate. Since MCQ results show models have basic TOC knowledge, we restrict LLM judgment solely to assessing Completeness and Correctness under a fixed rubric. The Effectiveness score does not use LLMs. It uses vector matching to check method suitability, rule-based checks for compatibility, and equations for transparency and timing. All parameters for these checks come directly from our knowledge base, daily lab records, over 300 quantitative experiments, and the literature excerpts.

Human evaluation (R1, R2, R3). We used a two-stage expert-panel procedure. First, one wet-lab TOC expert blindly scored protocols using the CCE rubric. Then, a panel of five TOC wet-lab experts and one knowledge-graph expert reviewed and adjusted these scores through discussion. They combined their different backgrounds to judge complex cases such

as incubation-time scaling and fluorescence compatibility. This consensus approach ensures consistency and addresses the limitations of single-rater or independent multi-rater voting. As shown in Figure 3, across 260 protocols, the automated Completeness and Correctness scores correlated with expert scores at $r=0.77$ and $r=0.69$, respectively, while the Effectiveness metric achieved $r=0.83$.

Baseline ablation & reproducibility (R2). To address the reviewer's concern, we added a lightweight ablation study on 20 OEQs, testing both GPT-5.2-Fast and Qwen3-Max. The RAG setup used cached domain knowledge including method-related text and compatibility records. For GPT-5.2-Fast, score improved from 66.4 to 69.7 with RAG; for Qwen3-Max, it improved from 62.2 to 66.3. Broader agentic baselines were not included because general tools like web search and code execution do not support TOC-specific protocol validation. Agentic and tool-augmented protocol design will be addressed in future extensions. Finally, The repository, temporarily offline for anonymity, is now accessible again.

2. I hereby, on behalf of all co-authors, give permission to make this response public
Agreement accepted
